Linear and Matrix Transformations

Recall from last class:

- Define $T : \mathbb{R}^n \to \mathbb{R}^m$ by $T(\vec{x}) = A\vec{x}$ where A is an m by n matrix. Recall we call $\mathbb{R}^n$ the domain and $\mathbb{R}^m$ the codmain. The set of outputs, $\{T(\vec{x}) : \vec{x} \in \mathbb{R}^n\}$, is the image or the range of T.
- $T : \mathbb{R}^n \to \mathbb{R}^m$ is **linear** if
  1. $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v}$ in $\mathbb{R}^n$,
  2. $T(c\vec{u}) = cT(\vec{u})$ for all $\vec{u}$ in $\mathbb{R}^n$ and c in $\mathbb{R}$.
- **Properties of Linear Transformations.**
  1. If T is linear, then $T(\vec{0}) = \vec{0}$.
  2. If T is linear, then $T(c\vec{u} + d\vec{v}) = cT(\vec{u}) + dT(\vec{v})$.
- All matrix transformations are linear.
- If $T : \mathbb{R}^n \to \mathbb{R}^m$ is given by $T(\vec{x}) = A\vec{x}$, then the image of T is the span of the columns of A; namely,
$$\mathrm{Im}(T) = \mathrm{Span}\{\vec{a}_1, \ldots, \vec{a}_n\}.$$
- We also saw various geometric transformations, including projections, dilations, and sheers.

1. Suppose $T : \mathbb{R}^2 \to \mathbb{R}^3$ is linear and

$$T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}5\\-7\\2\end{bmatrix}, \text{ and } T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}-3\\8\\0\end{bmatrix}.$$

What is

$$T\left(\begin{bmatrix}a\\b\end{bmatrix}\right)?$$

Theorem 10: Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear. Then there exists a unique matrix A such that $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$.

2. Prove the Theorem.

3. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation given by rotating $\vec{x}$ by θ counterclockwise. Find the associated matrix of T.

4. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be defined by $T(\vec{x}) = A\vec{x}$ where

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

Is T injective and/or surjective?

Theorem 11: Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear. Then T is injective if and only if $T(\vec{x}) = \vec{0}$ has only the trivial solution.

5. Prove the theorem.

Theorem 12: Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear, and let A be the associated matrix. Then

(a) T is surjective if and only if the columns of A span $\mathbb{R}^m$.

(b) T is injective if and only if the columns of A are linearly independent.

6. Prove the theorem.